Zakhar Yagudin

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SUMMARY

Lead Robotics / Perception Engineer with **5+ years** building production 3D detection, sensor-fusion, and control stacks for autonomous trucks and racing platforms. Shipped **one of the first L4 autonomous freight deployments in Europe** (Ozon) and led the team that took **2nd place at A2RL Silver League 2025**, the world's premier autonomous-racing championship. **7 publications** at IEEE including ICRA, IROS, IV and arXiv (3 first-author, 75+ citations) on VLM-based driving, multi-sensor 3D detection, and end-to-end autonomy. Currently a PhD candidate at Skoltech working on diffusion-based forecasting and deep RL for high-speed racing, while leading a 10-engineer perception–planning–control team.

EXPERIENCE

Lead Computer Vision Engineer — Khalifa University, Team FlyEagle Jan 2025 – Present *(full-time)*
Autonomous racing — A2RL championship; perception, planning, control *Abu Dhabi, UAE*

- Led a **10-engineer team** building the full autonomy stack for an A2RL race car at **270+ km/h**; owned architecture, roadmap, sprint decomposition, code review, and CI/CD.
- Secured **2nd place at A2RL Silver League 2025** against **12 international university teams** by shipping a **30 FPS** camera+LiDAR 3D-detection pipeline with **<40 ms end-to-end latency** on Jetson AGX Orin.
- Authored **3 publications** on real-time 3D detection and cross-domain racing perception; established lab's evaluation harness and reproducibility standards.

Computer Vision Researcher — 168ROBOTICS Jun 2024 – Present *(part-time)*
Multi-sensor 3D perception for autonomous platforms *Moscow, Russia*

- Designed and shipped a **LiDAR 3D-detection baseline from scratch**, reaching **NDS 80** on a **15k-frame internal dataset**; defined the data pipeline, model architecture, and evaluation protocol.
- Reduced inference latency **1.5×** via TensorRT INT8 quantization and CUDA-kernel tuning, enabling **90 FPS deployment on Jetson Orin**.

Software Engineer, Autonomous Driving — Ozon Tech Jan 2021 – May 2024 *(full-time)*
L4 autonomous freight trucks (MAN TGM, Volvo FH) *Innopolis, Russia*

- Owned the **lateral and longitudinal control module** (C++17) for MAN TGM and Volvo FH autonomous trucks; ran on STM32-based ECU at ∼**100 Hz** control loop.
- Achieved ∼**30 cm lateral tracking error at 80 km/h** on M-7 highway public-road trials; tuned MPC + CLF/CBF safety layer for high-mass dynamics.
- Shipped controller to revenue trucks logging **100+ km of autonomous miles on the M-7 corridor** — contributing to **one of the first L4 freight deployments in Europe**.

Founder & CEO — StraightenUp Sep 2024 – Present *(part-time)*
AI-powered wearable for posture monitoring — Skolkovo resident *Moscow / Skolkovo* Co-founded HealthTech wearable; defined product, R&D, and GTM. Secured Skolkovo residency; semifinalist at Sevan Startup Summit 2025.

Co-founder / CV Engineer — Chain of Blocks 2024 – 2025 *(acquired)*
Mobile App based on Smart contracts in TON *Russia* Founded Blockchain company **acquired in 2025**.

SELECTED PUBLICATIONS

Full list (7 papers, 3 first-author, 75+ citations) on Google Scholar.

- **Yagudin, Z.**, Tananakin, M., Guo, Z., Madjid, N.A., Mebrahtu, M., Dias, J., Khonji, M., Tsetserukou, D. *PerspectiveFusion: A Lightweight Real-time 3D Detector for Autonomous Driving.* **IROS 2025**.
- **Yagudin, Z.**, Guo, Z., Lykov, A., Konenkov, M., Tsetserukou, D. *VLM-Auto: VLM-based Autonomous Driving Assistant with Human-like Behavior and Understanding for Complex Road Scenes.* arXiv:2405.05885, 2024 (**57 citations**).
- **Yagudin, Z.**, Mebrahtu, M., Jin, R., Huang, J., Yue, Y., Tsetserukou, D., Dias, J., et al. *EagleVision: A Multi-Task Benchmark for Cross-Domain Perception in High-Speed Autonomous Racing.* arXiv:2604.11400, 2026.
- Guo, Z., Lin, X., **Yagudin, Z.**, Lykov, A., Wang, Y., Li, Y., Tsetserukou, D. *METDrive: Multimodal End-to-End Autonomous Driving with Temporal Guidance.* **ICRA 2025**.

EDUCATION

Skoltech — PhD Candidate, Robotics 2025 – Present · Moscow, Russia
Topic: Uncertainty-Aware Trajectory Planning in Autonomous Racing via Diffusion-Based Opponent Forecasting and Deep Reinforcement Learning.

Skoltech — MSc, Robotics 2023 – 2025 · Moscow, Russia
GPA 5.0/5.0. Focus: Autonomous Driving, Multi-modal Perception, Deep Learning. Thesis: 3D Object Detection using Vision Transformers and Sensor Fusion.

Innopolis University — BSc, Computer Science (Robotics) 2019 – 2023 · Innopolis, Russia
GPA 4.6/5.0. Thesis: Controller design for autonomous vehicles (MPC, CLF/CBF).

SKILLS

Languages: C/C++ (C++17), Python, CUDA.
DL & Deployment: PyTorch, TensorRT, ONNX, Hugging Face; INT8 quantization, real-time inference on Jetson Orin / STM32.
Perception: 3D Object Detection, Camera+LiDAR Sensor Fusion, BEV, Vision Transformers, VLM / VLA, multi-task perception.
Robotics & Control: ROS, MPC, CLF/CBF, behavior cloning, PPO/SAC, sim2real; Isaac Lab, MuJoCo, CARLA.
Leadership: Technical lead, roadmap & architecture ownership, hiring, code review, cross-functional alignment, founder experience.